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Predicting the Outcome of a Stolen Base

1.1 Background

Stolen bases surged in Major League Baseball following the introduction of new rules in 2023. Players stole 3,503 bases that season, a dramatic increase from 2,486 in 2022 (Harrigan). In the following season, the total climbed to 3,617 stolen bases, the most in 109 years and the third-highest since 1900 (Schoenfield).

This rise was driven by several rule changes enforced before the 2023 season. These included the pitch clock (which limited how long pitchers could hold the ball), restrictions on pickoff attempts, and larger bases (which shortened the distance between bases by about 4.5 inches). With these changes, I set out to build a program that predicts whether a baserunner will successfully steal second base, using the variables related to the pitcher, catcher, and the runner himself.

1.2 Methodology

To predict the stolen base outcome, I adopted a mathematical approach that compares the time it takes for the runner to reach second base with the time it takes for the ball to arrive there. The program accepts four inputs: pitch velocity, baserunner, baserunner's jump, and catcher. By contrasting the runner's time with the defensive time, the model generates a binary comparison between both values, predicting either a successful or unsuccessful stolen base attempt.

Ball to Second Timing

The first step was to calculate the time required for the ball to travel from the pitcher's hand to second base. This total defensive time is the sum of two components:

- **Pitch travel time:** The user inputted pitch velocity is converted into a time interval for the ball to cover the distance from the pitcher's hand to home plate (60.6 feet).
- **Catcher's pop time:** The average competitive pop time for the selected catcher, representing the transfer and throw to second base.

An important assumption made here is that all pitchers release the ball at 60.6 feet. The model also assumes that the throw to second base is perfectly accurate and that no additional time is required for the second baseman or shortstop to apply the tag to the runner.

Together, these values represent the defense's ability to deliver the ball to second base.

Runner Timing

The offensive side of the calculation focuses on the baserunner's jump and sprint speed.

- **Jump definition:** The jump is the distance the runner gains off first base at pitch release.
- **Jump calculation:** Using historical stolen base data, the minimum and maximum jump distances are identified. The user inputted jump (scaled 1–10) is mapped within this range:

$$new_jump = (jump_range \times \frac{jump}{10}) + min_jump$$

where *jump_range* represents the distance the baserunner was able to gain off first base.

- **Remaining distance:** Subtracting the jump from 60 feet yields the distance left to cover.
- **Time estimate:** The runner's time is calculated either by applying sprint speed to the remaining distance or by using a trendline derived from split times over 0–90 feet.

Another assumption is that the sprint speed measured from home to first is equivalent to the sprint speed when stealing second base.

Analytical Comparison

Once both values are computed, the model performs a direct comparison:

- If **Runner Time** < **Ball Time**, the prediction favors a successful stolen base.
- If **Runner Time** $\geq$ **Ball Time**, the prediction favors an unsuccessful attempt.

This binary framework provides a clear mathematical method for evaluating stolen base outcomes.

1.3 Conclusion

This project provided a valuable introduction to applying data science concepts to baseball analytics. It established a mathematical framework for evaluating stolen base attempts by comparing runner and ball timing. While the current model relies on simplifying assumptions and limited data, it demonstrates how teams could use this predictor in their scouting. From an offensive perspective, teams could leverage the model to identify favorable baserunning opportunities when a runner has a high probability of successfully stealing second against a given pitcher + catcher combination. From a defensive perspective, teams could use the predictor

to manage the running game, for example, by reducing a pitcher's time to the plate or improving a catcher's pop time.

Future Considerations

Future iterations of this project should incorporate additional variables that influence stolen base outcomes. Key factors include the pitch's location to home plate, the accuracy and quality of the throw to second base, and the pitcher's release distance relative to the plate.

Several assumptions in the current model also require refinement:

- **Pitcher release point:** The model assumes a uniform release distance of 60.6 feet, though in reality release points vary by height, mechanics, and delivery style. For example, a 95 mph fastball from Sonny Gray may reach home plate later than the same pitch from Chris Sale due to differences in release distance.
- **Sprint speed equivalence:** The model equates sprint speed from home to first with sprint speed when stealing second. While simplifying calculations, this overlooks differences in acceleration, effort, and situational context.
- **Throw accuracy and tag time:** The model assumes perfectly accurate throws and instantaneous tags. In reality, both introduce variability that can add crucial milliseconds, often determining whether the runner is safe at second base.

Acknowledging these limitations not only highlights weaknesses in the current framework but also points directly to opportunities for refinement in future work.

With richer datasets, these limitations could be addressed by integrating advanced statistical techniques such as logistic regression. This approach would allow the model to train

on test data and generate probabilistic predictions rather than binary outcomes. Such refinements would move the project beyond simplified equations toward a comprehensive predictive framework capable of capturing the full complexity of stolen base attempts.

1.4 Works cited

Harrigan, Thomas. “MLB Records Its Highest Steals Total in Season Since 1915.” *MLB.com*, 2 Oct. 2024.

Schoenfield, David. “‘It Gives Energy to the Whole Field’: MLB’s Stolen Base Resurgence.” *ESPN*, 2 Oct. 2024,